



Qualification Pack



Mechanic High Speed Rotating Machinery (Steel Plant)

QP Code: ISC/Q0916

Version: 1.0

NSQF Level: 4

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ISC/Q0916: Mechanic High Speed Rotating Machinery (Steel Plant)

Brief Job Description

The individual in this role performs mechanical maintenance, of fans, compressors, turbo blowers, sinter fans etc. by following organizational specified procedures and standards.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0953: Prepare for maintenance of high-speed rotating machinery](#)
3. [ISC/N0954: Perform maintenance of high-speed rotating machinery](#)
4. [ISC/N0955: Perform high speed rotating machinery post-maintenance activities](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Sponge Iron, Steel, Ferro Alloys, Re-Rollers, Refractory, Mechanical Maintenance
Occupation	Mechanical Maintenance
Country	India
NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8131.15



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Minimum Educational Qualification & Experience	12th grade Pass (In science) with 6 Months of experience In relevant field OR 11th Class (In science) with 1.5 years of experience In relevant field OR 10th Class with 3 Years of experience in relevant field OR 10th Class (2 years of NTC / NAC / CTS in Fitter, Machinist, Electrical) OR Pursuing 2nd year of 3-year regular Diploma (after 10th) (in Metallurgy / Mechanical / Production / Electrical)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-04-IS-03630-2025-V1-IISSSC
NQR Version	1



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ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Use protective clothing/equipment for specific tasks and work conditions
Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks
Glasses/gloves/visors
Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator

Health and safety procedures

To be competent, the user/individual on the job must be able to:

- PC2.** State the name and location of people responsible for health and safety in the workplace
Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.
- PC3.** State the names and location of documents that refer to health and safety in the workplace
- PC4.** Identify job-site hazardous work and state possible causes of risk or accident in the workplace
Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.
- PC5.** Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role
Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors
- PC6.** State location of general health and safety equipment in the workplace



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- PC7.** Inspect for faults, set up and safely use steps and ladders in general use
Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.
Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.
- PC8.** Work safely in and around trenches, elevated places and confined areas
- PC9.** Lift heavy objects safely using correct procedures
- PC10.** Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces
- PC11.** Identify common hazard signs displayed in various areas
- PC12.** Retrieve and/or point out documents that refer to health and safety in the workplace

Fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC13.** Use the various appropriate fire extinguishers on different types of fires correctly.
Fire extinguishers: Sand Water Foam Co2 Dry powder
Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents)
Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.
- PC14.** Demonstrate rescue techniques applied during fire hazard
- PC15.** Demonstrate good housekeeping in order to prevent fire hazards
- PC16.** Demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC17.** Demonstrate how to free a person from electrocution
- PC18.** Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC19.** Demonstrate basic techniques of bandaging
- PC20.** Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below :
Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.
- PC21.** Perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC22.** Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC23.** Demonstrate the artificial respiration and the CPR Process
- PC24.** Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work



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- PC25.** Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer
- PC26.** Demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace
- KU2.** State the names and location of documents that refer to health and safety in the workplace
- KU3.** Meaning of hazards and risks
- KU4.** Health and safety hazards commonly present in the work environment and related precautions
- KU5.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** Activities and causes of risk and accident
- KU7.** Methods of accident prevention
- KU8.** Safe working practices when working with tools and machines
- KU9.** Safe working practices while working at various hazardous sites
- KU10.** Where to find all the general health and safety equipment in the workplace
- KU11.** Various dangers associated with the use of electrical equipment
- KU12.** Preventative and remedial actions to be taken in the case of exposure to toxic materials. Exposure: ingested, contact with skin, inhaled Preventative action: ventilation, masks, protective clothing/equipment Remedial action: immediate first aid, report to supervisor Materials: solvents, flux, lead
- KU13.** Importance of using protective clothing/equipment while working
- KU14.** Precautionary activities to prevent the fire accident Activities and causes: Physical actions Reading Listening to and giving instructions Inattention Sickness and incapacity (e.g. Drunkenness) Health hazards (e.g. Untreated injuries and contagious illness)
- KU15.** Various causes of fire
- KU16.** Techniques of using the different fire extinguishers
- KU17.** Different methods of extinguishing fire
- KU18.** Rescue techniques applied during a fire hazard
- KU19.** Various types of safety signs and what they mean
- KU20.** Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU21.** Content of written accident report



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- KU22.** Potential injuries and ill health associated with incorrect manual handling
- KU23.** Safe lifting and carrying practices
- KU24.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU25.** Potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Give clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS16.** Report problems that you cannot resolve to appropriate authority
- GS17.** Identify cause and effect relations in their area of work
- GS18.** Use cause and effect relations to anticipate potential problems and their solution



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	5	5	-	-
PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visors Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator	5	5	-	-
<i>Health and safety procedures</i>	15	40	-	-
PC2. State the name and location of people responsible for health and safety in the workplace Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.	-	4	-	-
PC3. State the names and location of documents that refer to health and safety in the workplace	-	1	-	-
PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors	5	5	-	-
PC6. State location of general health and safety equipment in the workplace	-	5	-	-
PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc. Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.	-	5	-	-
PC8. Work safely in and around trenches, elevated places and confined areas	-	4	-	-
PC9. Lift heavy objects safely using correct procedures	-	4	-	-
PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces	-	1	-	-
PC11. Identify common hazard signs displayed in various areas	5	1	-	-
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	-	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fire safety procedures</i>	10	15	-	-
PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: Sand Water Foam Co2 Dry powder Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.	5	5	-	-
PC14. Demonstrate rescue techniques applied during fire hazard	5	5	-	-
PC15. Demonstrate good housekeeping in order to prevent fire hazards	-	1	-	-
PC16. Demonstrate the correct use of a fire extinguisher	-	4	-	-
<i>Emergencies, rescue and first-aid procedures</i>	15	45	-	-
PC17. Demonstrate how to free a person from electrocution	-	5	-	-
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	5	-	-
PC19. Demonstrate basic techniques of bandaging	-	5	-	-
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments	-	5	-	-
PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	-	5	-	-
PC23. Demonstrate the artificial respiration and the CPR Process	-	5	-	-
PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work	-	4	-	-
PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer	5	5	-	-
PC26. Demonstrate correct method to move injured people and others during an emergency	-	1	-	-
NOS Total	45	105	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0953: Prepare for maintenance of high-speed rotating machinery

Description

This OS unit is about preparing for maintenance of maintenance of high-speed rotating machinery like fans, compressors, turbo blowers, sinter fans etc.

Scope

The scope covers the following :

- The scope covers the following:
- . Identify work requirements
- . Prepare for maintenance activities

Elements and Performance Criteria

Identify work requirements

To be competent, the user/individual on the job must be able to:

- PC1.** identify the machine, its manufacturer's specifications and functioning from the user manual
- PC2.** identify the work to be done by interpreting the checklist, engineering drawings, work order and SOPs etc.
- PC3.** interpret information from sections, views and scale of measurement used in drawing of the machine
- PC4.** identify correct limits, tolerance and fits of rotating machine by interpreting the drawings properly
- PC5.** identify the tools, equipment, consumables and spare parts required during the task

Prepare for maintenance activities

To be competent, the user/individual on the job must be able to:

- PC6.** use appropriate Personal Protective Equipment (PPE) for safe working in workshop
- PC7.** read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance
- PC8.** select and arrange the right tools, equipment, consumables and spare parts as per the SOP and job requirements
- PC9.** check the tools, equipment, consumables and spare parts for any defects and that they are as per the required quality standards
- PC10.** check and calibrate the tools and equipment before use
- PC11.** prepare tools, tackles, spares, lifting equipment etc. as per SOP/WI
- PC12.** use checklist and coordinate with the superior for confirming maintenance tasks
- PC13.** report damaged / defective components of equipment and tools as per the escalation matrix

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** company's maintenance standards and processes
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** working principles of high-speed rotating machinery like fans, compressors, turbo blowers, sinter fans etc. for successful operation of plant and piping systems
- KU4.** functioning and use of associated systems, accessories and components
- KU5.** limits, fits and tolerances of the rotating equipment
- KU6.** how to read engineering drawing and sketches of the equipment
- KU7.** Standard Operating Procedures (SOP) recommended by OEM for using tools, tackles, equipment
- KU8.** how to collect and store consumables, spare parts, tools etc. as per organizational procedures
- KU9.** normal running characteristics of rotating equipment
- KU10.** implications of not adhering to sequence of activities and operations
- KU11.** use of appropriate PPE, material handling equipment and tools for completing the maintenance tasks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify work requirements</i>	9	15	-	7
PC1. identify the machine, its manufacturer's specifications and functioning from the user manual	1	2	-	1
PC2. identify the work to be done by interpreting the checklist, engineering drawings, work order and SOPs etc.	1	2	-	1
PC3. interpret information from sections, views and scale of measurement used in drawing of the machine	3	5	-	1
PC4. identify correct limits, tolerance and fits of rotating machine by interpreting the drawings properly	2	3	-	2
PC5. identify the tools, equipment, consumables and spare parts required during the task	2	3	-	2
<i>Prepare for maintenance activities</i>	21	35	-	13
PC6. use appropriate Personal Protective Equipment (PPE) for safe working in workshop	1	2	-	-
PC7. read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance	2	3	-	2
PC8. select and arrange the right tools, equipment, consumables and spare parts as per the SOP and job requirements	2	3	-	1
PC9. check the tools, equipment, consumables and spare parts for any defects and that they are as per the required quality standards	4	7	-	3
PC10. check and calibrate the tools and equipment before use	4	7	-	2
PC11. prepare tools, tackles, spares, lifting equipment etc. as per SOP/WI	4	7	-	3



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use checklist and coordinate with the superior for confirming maintenance tasks	2	3	-	1
PC13. report damaged / defective components of equipment and tools as per the escalation matrix	2	3	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0953
NOS Name	Prepare for maintenance of high-speed rotating machinery
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

ISC/N0954: Perform maintenance of high-speed rotating machinery

Description

This OS unit is about performing the maintenance of high-speed rotating machinery as per the SOP.

Scope

The scope covers the following :

- The scope covers the following:
- Perform maintenance of high-speed rotating machinery

Elements and Performance Criteria

Perform maintenance of high-speed rotating machinery

To be competent, the user/individual on the job must be able to:

- PC1.** enquire with the user/operator about any problems/unusual conditions noticed on the rotating machinery
- PC2.** follow safety practices during maintenance activities as per organisational SOP
- PC3.** perform basic health check-up of rotating machinery as specified in the maintenance checklist
- PC4.** conduct visual and dimensional inspections to assess the components
- PC5.** check the equipment for any operational deviations, geometric inaccuracies or internal conditions to test its working status and expected conditions
- PC6.** perform various calculations to size and assess rotating equipment performance (head, discharge temp, inlet capacity, volumetric efficiency etc.)
- PC7.** check the machine parts for any wear and tear, damage, improper functioning etc.
- PC8.** dismantle the equipment and replace/change the spare parts and consumables as per the schedule
- PC9.** conduct breakdown maintenance and then inspect the machine for leakages, breakages, unusual noise, changes in noise, temperatures, pressures, flows, or vibration levels, under-rated performance or any other problem as reported in the complaint
- PC10.** use appropriate vibration and temperature measuring instruments to monitor the vibration and temperature in the equipment
- PC11.** read the gauges and record the observations at prescribed time intervals
- PC12.** analyse the readings to identify any deviations and correct them if not within the desired levels
- PC13.** check pipe work, small bore tubing and flexible hose for any defects
- PC14.** analyze the problems in operation related to packing, mechanical seals, bearings and flow control
- PC15.** repair or replace damaged components as per the requirement
- PC16.** use recommended consumables, tools and equipment for service and repair of equipment as per Standard Operating Procedures (SOP)
- PC17.** add lubricants and coolants into moving parts of machine as per SOP/WI



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- PC18.** assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel
- PC19.** dispose off waste materials such as failed parts/aggregates, as per organisation's policies

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's maintenance standards and processes
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** how to read maintenance schedules and checklists recommended by the machine manufacturer
- KU4.** various rotating equipment construction details
- KU5.** about equipment control systems (mechanical, Pneumatic, Hydraulic)
- KU6.** how to read and interpret pump and compressor curves
- KU7.** optimize the operation efficiency of machines
- KU8.** what affects the selection and design of rotating equipment (process conditions, fluid characteristics, interstaging, foundation needs etc.)
- KU9.** various calculations to size and assess rotating equipment performance (head, discharge temp, inlet capacity, volumetric efficiency etc.)
- KU10.** various problems related to packing, mechanical seals, bearings and flow control
- KU11.** best practices and guidelines for maintenance, problem solving and troubleshooting of rotating equipment
- KU12.** useful charts and diagrams to solve daily rotating equipment problems
- KU13.** corrective actions for common faults and failures in machine and its components
- KU14.** use of appropriate PPE, material handling equipment and tools for completing the maintenance tasks
- KU15.** documentation required regarding repair, maintenance and service performed

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read signals, work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision



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GS9. report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform maintenance of high-speed rotating machinery</i>	30	50	-	20
PC1. enquire with the user/operator about any problems/unusual conditions noticed on the rotating machinery	1	2	-	1
PC2. follow safety practices during maintenance activities as per organisational SOP	1	1	-	-
PC3. perform basic health check-up of rotating machinery as specified in the maintenance checklist	2	4	-	2
PC4. conduct visual and dimensional inspections to assess the components	2	3	-	1
PC5. check the equipment for any operational deviations, geometric inaccuracies or internal conditions to test its working status and expected conditions	2	3	-	2
PC6. perform various calculations to size and assess rotating equipment performance (head, discharge temp, inlet capacity, volumetric efficiency etc.)	2	3	-	1
PC7. check the machine parts for any wear and tear, damage, improper functioning etc.	2	3	-	1
PC8. dismantle the equipment and replace/change the spare parts and consumables as per the schedule	2	3	-	1
PC9. conduct breakdown maintenance and then inspect the machine for leakages, breakages, unusual noise, changes in noise, temperatures, pressures, flows, or vibration levels, under-rated performance or any other problem as reported in the complaint	3	5	-	2
PC10. use appropriate vibration and temperature measuring instruments to monitor the vibration and temperature in the equipment	2	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. read the gauges and record the observations at prescribed time intervals	1	2	-	1
PC12. analyse the readings to identify any deviations and correct them if not within the desired levels	1	2	-	1
PC13. check pipe work, small bore tubing and flexible hose for any defects	1	3	-	1
PC14. analyze the problems in operation related to packing, mechanical seals, bearings and flow control	2	3	-	1
PC15. repair or replace damaged components as per the requirement	2	3	-	1
PC16. use recommended consumables, tools and equipment for service and repair of equipment as per Standard Operating Procedures (SOP)	1	2	-	1
PC17. add lubricants and coolants into moving parts of machine as per SOP/WI	1	2	-	1
PC18. assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel	1	2	-	1
PC19. dispose off waste materials such as failed parts/aggregates, as per organisation's policies	1	1	-	-
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0954
NOS Name	Perform maintenance of high-speed rotating machinery
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

ISC/N0955: Perform high speed rotating machinery post-maintenance activities

Description

This OS unit is about performing post-maintenance activities such as trials, storage, housekeeping etc.

Scope

The scope covers the following :

- The scope covers the following:
- . Conducting trials and documentation
- . Perform post-maintenance activities

Elements and Performance Criteria

Conducting trials and documentation

To be competent, the user/individual on the job must be able to:

- PC1.** assemble back the covers, guards, clamps, insulation etc. of the equipment and prepare it for conducting the trials
- PC2.** shift and install the machine for the next procedure and record all repairs carried out, time taken and unplanned tasks encountered in the activity
- PC3.** check the machine/electrical panels component as per the control plan, WI for proper working
- PC4.** check vibrations in the machine to ensure they are within desired limits
- PC5.** conduct trials of the machine and verify that specified parameters are attained with no abnormalities
- PC6.** run few cycles of equipment and observe its functioning in the presence of operator
- PC7.** ensure that rotating equipment and auxiliary systems are operating within the OEM operating limits
- PC8.** change the maintenance due/status sticker
- PC9.** ensure that all maintenance points are adequately considered and report the superiors about maintenance activity done
- PC10.** record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities

Perform post-maintenance activities

To be competent, the user/individual on the job must be able to:

- PC11.** clean and store all the tools and equipment after completion of work
- PC12.** clean the work area after completion of work
- PC13.** remove the tools and equipment being from the work site before handing over
- PC14.** inform supervisor and concerned person regarding the job completion

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required
- KU3.** quality checks to be done and importance of the same
- KU4.** how to conduct trial of the equipment
- KU5.** post-maintenance processes like inspection, cleaning, storage etc.
- KU6.** methods of storage and cleaning tools and equipment
- KU7.** documentation required regarding repair, maintenance and service performed

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conducting trials and documentation</i>	25	41	-	17
PC1. assemble back the covers, guards, clamps, insulation etc. of the equipment and prepare it for conducting the trials	3	5	-	2
PC2. shift and install the machine for the next procedure and record all repairs carried out, time taken and unplanned tasks encountered in the activity	3	5	-	2
PC3. check the machine/electrical panels component as per the control plan, WI for proper working	3	5	-	2
PC4. check vibrations in the machine to ensure they are within desired limits	3	5	-	2
PC5. conduct trials of the machine and verify that specified parameters are attained with no abnormalities	3	5	-	2
PC6. run few cycles of equipment and observe its functioning in the presence of operator	3	5	-	2
PC7. ensure that rotating equipment and auxiliary systems are operating within the OEM operating limits	2	3	-	2
PC8. change the maintenance due/status sticker	2	3	-	1
PC9. ensure that all maintenance points are adequately considered and report the superiors about maintenance activity done	2	3	-	1
PC10. record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities	1	2	-	1
<i>Perform post-maintenance activities</i>	5	9	-	3
PC11. clean and store all the tools and equipment after completion of work	2	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. clean the work area after completion of work	1	2	-	1
PC13. remove the tools and equipment being from the work site before handing over	1	2	-	1
PC14. inform supervisor and concerned person regarding the job completion	1	2	-	-
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0955
NOS Name	Perform high speed rotating machinery post-maintenance activities
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQF Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70



Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	45	105	-	-	150	10
ISC/N0953.Prepare for maintenance of high-speed rotating machinery	30	50	0	20	100	20
ISC/N0954.Perform maintenance of high-speed rotating machinery	30	50	0	20	100	30
ISC/N0955.Perform high speed rotating machinery post-maintenance activities	30	50	0	20	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	155	285	-	60	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.